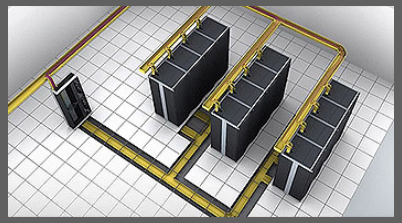
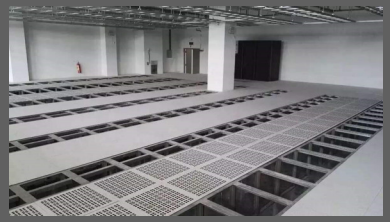
DATA CENTER

WHAT IS A DATA CENTER?

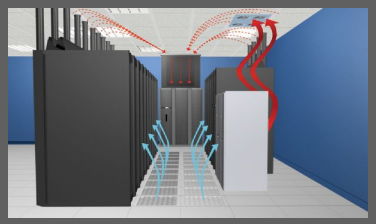
- A central location where all IT equipment and structured cable are housed
- It can be in many forms but will always have



Cable Management



Flooring



Cooling (HVAC)

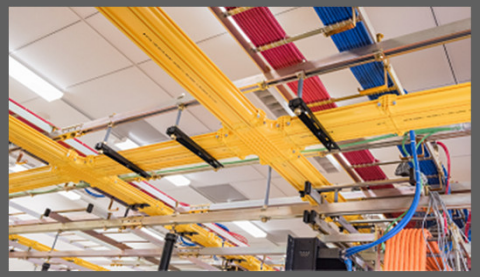
Equipment Rack

Power

DATA CENTER CABLING

CABLE MANAGEMENT

- Cable organization in a data center
- Cables are normally placed at the top of the room or under the floor
- Cables going into the data center can come from other
  - Rooms, buildings or outside the premise
- Cables can be color coded for certain purposes

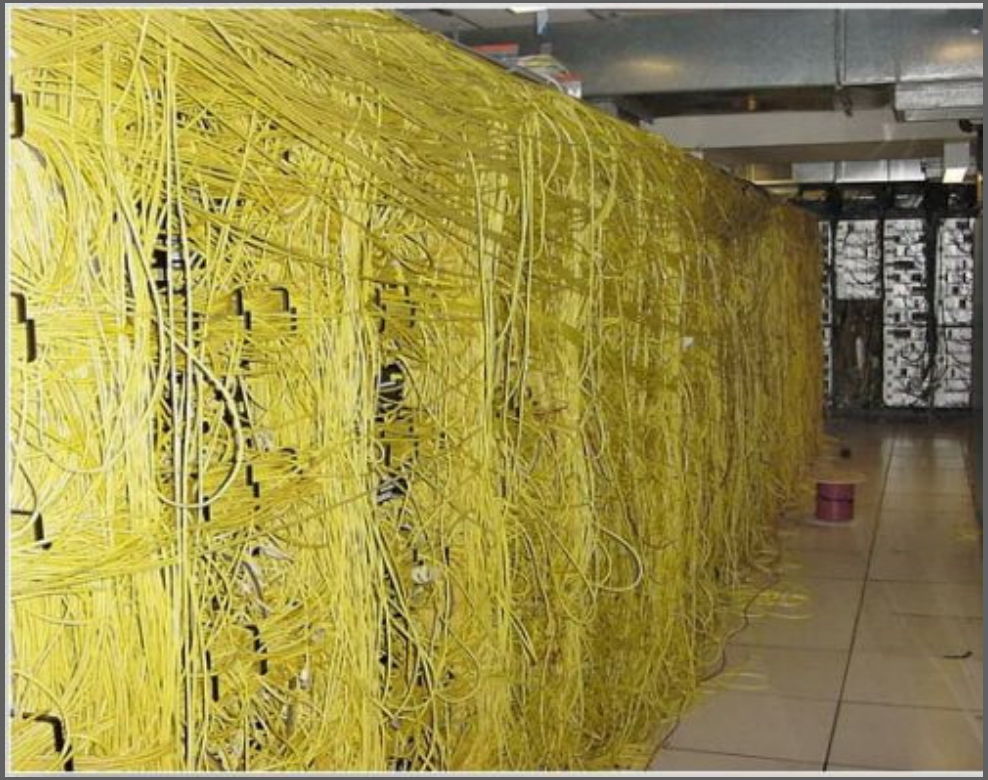


Overhead Cable Tray / Duct

Under Floor Cable Tray

Rack Cable Management

BAD CABLE MANAGEMENT PRACTICE



GOOD PRACTICE

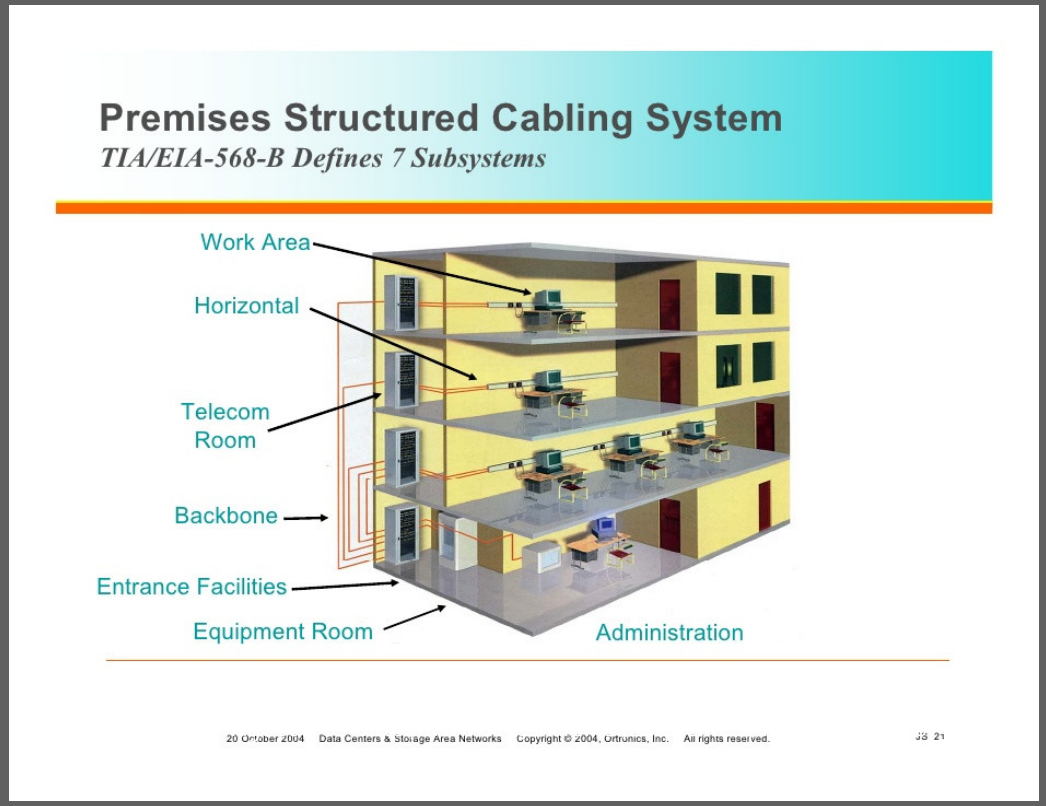
STRUCTURED CABLING

- The design and installation of a cabling system that will support multiple hardware uses
- Design covers from "work area" to "data center" cabling
- Depending on the country, there are standards for structured cabling
  - Europe – EN50173
  - US – EIA/TIA 568

STRUCTURED CABLING

- Cabling infrastructure consisting of standardized elements
- It has six subsystems
  - Entrance Facility
  - Equipment Room
  - Backbone Cabling
  - Horizontal Cabling
  - Telecommunication Room
  - Work Area

STRUCTURED CABLING



STRUCTURED CABLING SUBSYSTEMS

- Entrance Facility
  - Entrance facilities contain the cables, network demarcation point(s), connecting hardware, protection devices and other equipment that connect to the access provider (AP) or private network cabling
  - It includes connections between outside plant and inside building cabling
- Equipment Room
  - Environmentally controlled centralized space for telecommunications equipment
    - Usually more complex than a telecommunications room
  - Usually houses the main cross-connect
  - May also contain the intermediate cross-connects, horizontal cross-connects or both

Source - https://www.anixter.com/en_au/resources/literature/techbriefs/the-six-subsystems-of-a-structured-cabling-system.html

STRUCTURED CABLING SUBSYSTEMS

- Backbone Cabling
  - Provides interconnection between telecommunications rooms, equipment rooms, access provider spaces and entrance facilities
- Telecommunication Room
  - Houses the terminations of horizontal and backbone cables to connecting hardware including any jumpers or patch cords
  - It may also contain the intermediate or main cross-connect for different portions of the backbone cabling system
  - Also provides a controlled environment to house telecommunications equipment, connecting hardware and splice closures serving a portion of the building

Source - https://www.anixter.com/en_au/resources/literature/techbriefs/the-six-subsystems-of-a-structured-cabling-system.html

STRUCTURED CABLING SUBSYSTEMS

- Horizontal Cabling
  - Extends from the work area's telecommunications information outlet to the telecommunications room
  - It includes horizontal cable, mechanical terminations, jumpers and patch cords located in the telecommunications room
  - This is normally 90 meters from the work area to the telecommunications room
- Work Area
  - Components extend from the telecommunications outlet/connector end of the horizontal cabling system to the work area equipment
  - A minimum of two telecommunications outlets should be provided for each work area

Source - https://www.anixter.com/en_au/resources/literature/techbriefs/the-six-subsystems-of-a-structured-cabling-system.html

DATA CENTER FLOORING

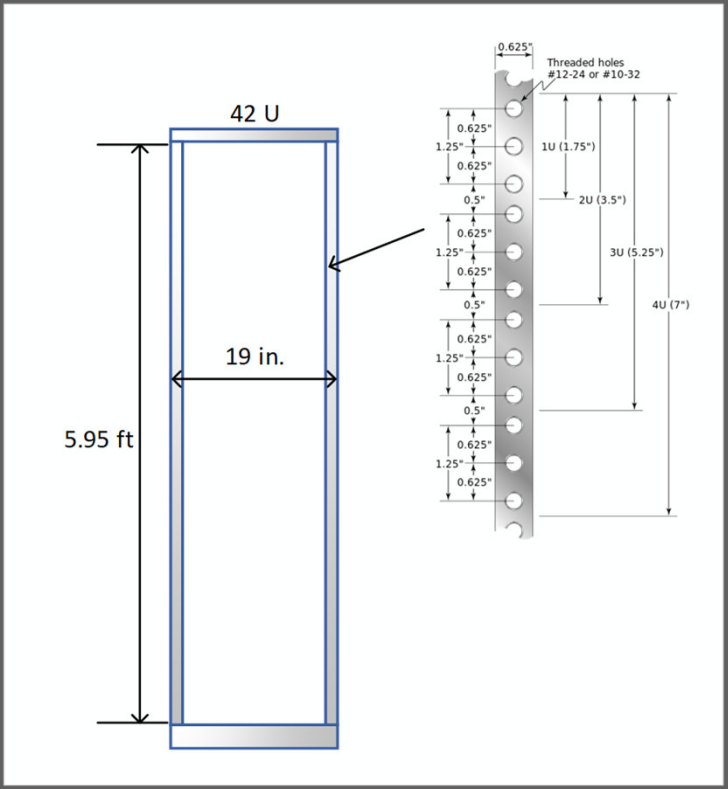
DATA CENTER FLOORING

- Data center uses a flooring system called “raised flooring”
- Raised flooring has a steel structure that should be able to carry the weight of the rack
- Flooring is modular and can be opened

EQUIPMENT RACK

EQUIPMENT RACK

- A physical steel frame that is designed to house servers, network equipment and cables
- A standard rack is 19 in. wide and 36 in. deep
- Height is specified as a "u"
  - A "u" is equivalent to 1.7 in. or 44.45 mm
  - Ex: if a rack is 42u this means that it has a height of 71.4 in. or 5.95 ft.
- Holes on the side of the rack is also standardized
  - Two adjacent holes are .625 in. apart from the middle
- Equipment uses bracket for mounting on the rack
  - Rack-type equipment are also uses "u" for the height

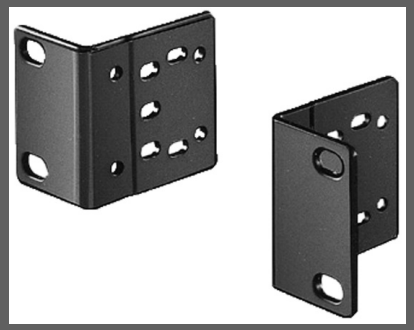


EXAMPLE EQUIPMENT RACK

EXAMPLE RACK-MOUNT EQUIPMENT

1u Rack-mount Server

2u Rack-mount Server



1u Bracket for Mounting

COOLING

COOLING

- Data centers need cooling as equipment in the rack produces heat
  - When the room is hot, equipment will not function properly or may overheat
- Cooling should also be energy efficient
  - Cooling can take up to 50% of the total power need to run the data center
- Data centers should have a temperature between 21c to 24c
- There are many types of cooling system for the data center

AIR CONDITIONING

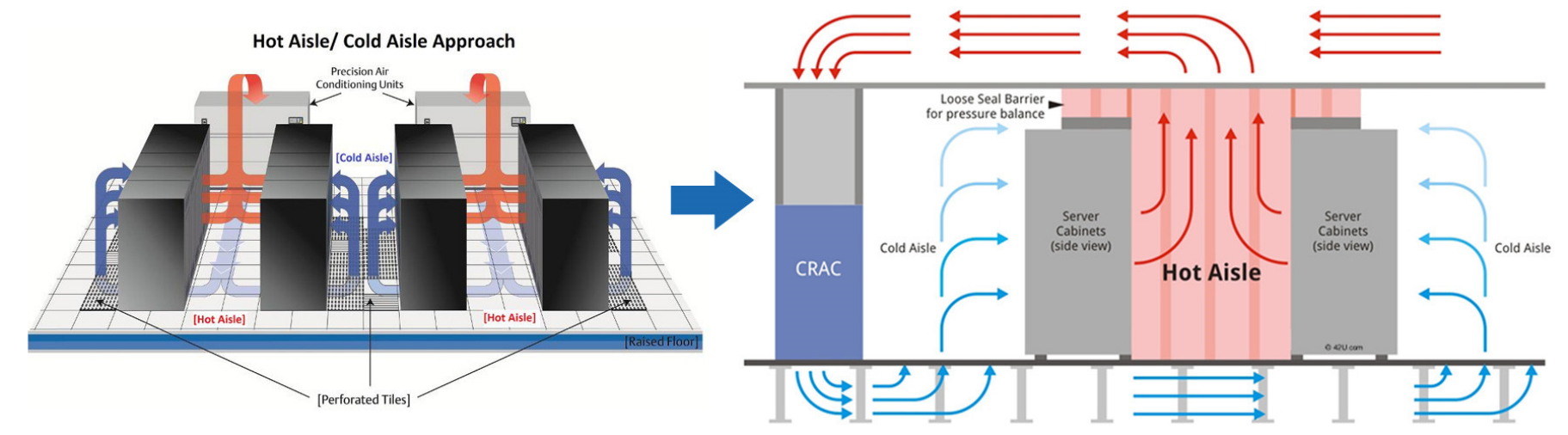
- Data center uses air conditioning called Precise Air Conditioning Unit
  - Controls the humidity and temperature precisely in the environment
- There are data centers that uses standard air conditioners
  - Also known as comfort air conditioners
- Precise air conditioning are better because it ensures the correct environment for the equipment

CONSIDERATIONS IN COOLING THE DATA CENTER

- Type of air conditioner
  - CRAC (Computer Room Air Conditioning) – uses refrigerant to cool the environment
  - CRAH (Computer Room Air Handler) – uses chilled water to cool the environment
- Cooling capacity of the air conditioner
  - Normally specified as BTU (British Thermal Units)
  - Computed by engineers guided by the heat given off by the equipment and people inside the room
- Rack placing
  - Placement of rack affects cooling
  - Normal design uses the hot aisle / cold aisle design

HOT AISLE / COLD AISLE

- One aisle has cold air and is vent into the rack
- All hot air from the equipment goes out in another aisle
- Hot air is then vented out of the data center



POWER

ELECTRICAL POWER

- Each equipment used in the data center has a rating in
  - Volts
  - Amperes
  - Power - Watts or VA (not the same as watts)
- Example:
  - A server can be rated as 220V / 3.5A or 220V / 600W
    - This means server runs 220Vac at 600W of power
- Data center supply is normally rated with current in Amperes
- Never go beyond the rated power supply of the data center
  - Will cause an overload
  - Also check the PDU's rating so as to exceed it

POWER REDUNDANCY

- Data centers can have electrical generators for power during outages
- Data centers also uses Uninterruptible Power Supply (UPS) during transition to generators in power outages
  - UPS are not designed for long power outages as it uses only battery to power equipment

Electrical Generator

Home / Personal UPS

High Capacity UPS
(For data centers)

Large Data Center UPS

SECURITY

SECURITY IN DATA CENTER

- Physical Security
  - Must ensure that only authorized personnel is inside the data center
  - Employment of locks or key card type locks
- Fire protection
  - Smoke detectors
  - Fire extinguishers
  - Fire suppression systems
- Safety
  - Fire exit

DATA CENTER